accessed December 23, 2016.

their views were not too far apart: Ibid.

"Quantum Theory and the Knowability of Nature": The original title, in German, was "Quantentheorie und Erkennbarkeit der Natural." Following William H. Werkmeister 1936, "The Second International Congress for the Unity of Science," *Philosophical Review* 45 (6): 593–600, in translating "Erkennbarkeit," in this context, as "knowability."

"neither true nor false, but *meaningless*": Ibid. Emphasis in original.

distasteful, such as vitalism: Abraham Pais 1991, *Niels Bohr's Times in Physics, Philosophy, and Polity* (Oxford University Press), p. 443.

"express[es] himself somewhat unclearly": Faye 2007.

"the sense of my efforts": Ibid.

"only when the experiment includes a position determination": Schiff 1955, *Quantum Mechanics*, 2nd ed. (McGraw-Hill), p. 6.

"there is no orbit in the ordinary sense": Heisenberg 1958, p. 48.

"misuse of language which… cannot be justified": Ibid.

couldn't actually justify most formulations of the Copenhagen interpretation: At least one prominent positivist, Hans Reichenbach, recognized that the verification theory of meaning couldn't be used to straightforwardly argue for the Copenhagen interpretation. "It would be wrong to argue that statements about the value of an entity before a measurement are meaningless because they are not verifiable. Statements about the value after the measurement are not verifiable either. If, in the [Bohr-Heisenberg interpretation], the one sort of statement are [*sic*] forbidden and the other admitted, this must be considered as a rule which, logically speaking, is arbitrary, and which can be judged only from the standpoint of expediency" (Reichenbach 1944, *Philosophic Foundations of Quantum Mechanics* [Dover], p. 142). Reichenbach dismissed the Copenhagen interpretation as problematic, because it promotes an ad hoc principle about what statements are meaningless into a law of physics. He argued instead for an interpretation based on a three-valued logic system, but this was later discovered to have its own problems related to the boundary between the microscopic and the macroscopic.

"suffered for National Socialism": Friedrich Stadler 2001, "Documentation: The Murder of Moritz Schlick," in *The Vienna Circle:*